FCE 15/50

High-Pressure Washer

User Manual

Contents

Warranty Information

Limited Warranty:

Fussen provides a 12-month or 1,000-hour (whichever comes first) warranty against material and manufacture defects for the company's high-pressure washers, provided the washer is used normally and in accordance with all operating instructions.
If the equipment is sold to an end user, the applicable warranty period begins on the date of delivery to the end user.

If the equipment is used for rental purposes, the applicable warranty period begins on the date the equipment is delivered to the party holding it for rent.

During the warranty period, this limited warranty may be transferred to any subsequent assignee.

This limited warranty is the sole and exclusive warranty provided by Fussen.

Repair:

If the equipment malfunctions within the warranty period and Fussen confirms through inspection that the fault is due to material defects or flaws in the original manufacturing process, Fussen is responsible to repair or replace the components.
Repairs or replacements shall be performed at Fussen's facility, the customer's location, or other sites approved by Fussen.

The aforementioned repair solution shall be the sole and exclusive remedy for any valid warranty claim submitted by either party.

Fussen's limited warranty does not apply to (and Fussen is not liable for):

- Major components and commercial accessories with warranties independent of the original manufacturer, such as engines, couplings, pressure gauges, and high-pressure hoses.
- Routine maintenance and servicing performed by the end user.
- Normal wear parts, such as couplings, belts, packing, seals, and pressure-bearing fittings.
- Malfunctions caused by not following Fussen's recommendations or due to human damage.
- Repairs, alterations, or modifications made without written permission from Fussen. Negative impact on the stability, operation, or reliability of the equipment designed and manufactured by the original manufacturer caused by the aforementioned actions.
- Misuse of parts, operational errors, accidents, or improper maintenance.

Notes

Using any parts not approved by Fussen may also void this warranty. Fussen reserves the right to determine at its sole discretion whether the warranty remains valid when unauthorized parts are used.
Fussen shall not be liable for any losses, injuries, or damages of any kind caused to any organization by any defect or malfunction of the equipment that is not listed in this warranty.
The essential warranty clause supersedes all other express or implied warranties, including but not limited to the merchantability of the equipment (the characteristics of the equipment that allow it to be sold at a reasonable price in a short period) or suitability for a particular purpose.

To avoid any doubt, Fussen shall not be liable for any indirect, special, incidental, or consequential damages, including but not limited to loss of use or lost profits.
Except for the written materials, catalogs, or specifications provided by Fussen with the equipment, Fussen does not claim any additional performance features for this equipment.
Fussen has not authorized any individual or affiliate to modify these warranty terms, nor has it provided any other warranties. Furthermore, Fussen has not authorized any other company to assume any responsibility related to the sale, service, or repair of any equipment manufactured by Fussen on its behalf.
If such behavior is identified, Fussen will take appropriate legal action within eighteen months. Fussen reserves the right to modify designs or improve products, but this does not obligate Fussen to make changes or improvements to previously manufactured products.

Precautions

Observe the following when using the equipment:

- Power

  Must be a 380V/15KW industrial power supply. If an extension cable is needed, consult an electrician, as the cross-sectional area of the cable must be proportional to the length of the extension.

- Water

  The water source should preferably be clean drinking tap water. The use of recycled water, industrial water, firefighting water, irrigation water, or well/river water is prohibited. Water flow rate must be greater than 1600 L/h. It is recommended to use a 3/4-inch steel wire-braided corrugated pipe for the water supply inlet.

- Avoid frequent equipment starts or frequent on and off of the gun. It is recommended to perform these operations no more than twice within one minute.
- Do not direct high-pressure water at electrical equipment, people, animals, or fragile items.
- Never point the nozzle at anyone. The equipment must be used and managed by designated personnel for specified purposes only.
- Check the filter element daily and replace it according to water quality conditions. The filter element must be replaced within 50 hours.
- If the high-pressure hose shows wear and exposes the steel wire, replace it immediately.
- Check the lubricant level and quality daily, and replace the lubricant at regular intervals.
- Crankcase oil change intervals: Replace after the first 100 hours of operation, then every 300 hours thereafter.
- If any part of the equipment leaks during operation, it must be addressed immediately to prevent damage to the pressure regulating valve.

1. Preface

The high-pressure cleaner is designed by a leading manufacturer with advanced technology in China.

To ensure the product's longevity and durability, we recommend carefully reading this operation manual before using the equipment.

This operation manual is provided for: operators, maintenance technicians, and repair technicians.

As our company continues to develop new products, updates to the content in this user guide may be made without prior notice.

2. Introduction

The FUSSEN series high-pressure washers allow you to achieve optimal cleaning results in less time.

FUSSEN provides you with high-performance equipment and accessories to meet your demanding and rigorous cleaning requirements.

The construction and design of the equipment, along with its powder-coated steel frame, provide optimal protection during transportation and use and ensures solutions to all challenges in high-pressure cleaning tasks.

The high-pressure cleaner series is designed to handle demanding cleaning tasks and withstand challenging environmental conditions.

All pumps and components in contact with water are constructed from corrosion-resistant materials, including ceramic pistons, long-life seals, and valves, ensuring the water gun's durability and longevity.

3. Technical Parameters

Item	Parameter
Model	FCE 15/50
Maximum Pressure (bar/MPa/psi)	0-500/0-50/0-7250
Maximum Flow (L/h)	900
Drive (Volts/Phase/Hz)	380/3/50
Power (kW)	15
Maximum inlet temperature (°C)	60
Speed (rpm)	1450
Weight (kg)	260
Dimensions (Length × Width × Height in mm)	1200×860×950

4. Applications

The high-pressure washer is capable of removing any type of dirt adhering to vehicles, machinery, ships, or buildings:

- Cement/Concrete
- Surface adherents/Surface treatment
- Paint and rust
- Caked-on dirt
- Optional accessories allow for additional tasks:
  - Sandblasting
  - Sewer cleaning

5. Safety Instructions

To prevent accidents and protect operators, bystanders, and surrounding machinery, place warning signs nearby or take safety measures. Regard the equipment as a high-speed cutting tool.

Do not allow children, individuals with behavioral disorders, or anyone who has not read the safety instructions to operate the equipment.

Do not use flammable, highly toxic, or explosive liquids as cleaning agents.

To avoid electric shock, do not use the equipment outdoors in the rain, touch the plug or socket with wet hands, or operate it with a damaged power cord.

The equipment must be connected to a properly grounded power supply. Risk of electric shock. Using a safety circuit breaker will provide additional protection for the operator (30 mA).

Choose the appropriate power cord and socket, and pay special attention to the grounding configuration. Connect only to a power supply with a grounded configuration. The equipment must be installed by a professional electrician.

To ensure the safety of cleaning personnel, inspect the accessories before use and check the insulation of equipment wiring. If any issues are detected, do not operate the equipment and perform troubleshooting immediately.

When using the equipment and its accessories, protect your eyes to avoid injury from flying debris.

When using the equipment, wear protective clothing and shoes. Do not point the high-pressure nozzle toward yourself. Do not use high-pressure water to wash clothing or footwear.

Bystanders must be kept away from the work area to prevent potential hazards.

Do not spray at people or animals, as high-pressure water can cause serious injury.

Do not spray water on live electrical devices or the equipment itself.

Ensure the equipment is turned off and the power cord is disconnected from the power source when repairing or maintaining the equipment, or when replacing parts.

Note: This equipment is to be used only within its designed scope. Any other use will be considered inappropriate and may pose a hazard. The manufacturer is not responsible for any improper or hazardous operation.

DO	DON'T
	

6. Cautions for Equipment Operation

- The user manual, as an integral part of the equipment, must be kept properly for future reference. If you sell the equipment, transfer the manual to the new owner.
- Before starting the equipment, ensure the water source is turned on and water is being supplied. Operating the equipment without water may damage the water seal.
- Do not unplug by pulling the cord.
- If you are too far from the object being cleaned, do not move the equipment by pulling on the high-pressure hose or power cord.
- Prevent the equipment from freezing in winter.
- Do not adjust the water flow angle while the equipment is running.
- The cross-sectional area of the extension cord must be suitable for its length. For example, the longer the extension cord, the larger its cross-sectional area:

25m extension cord cross-sectional area	50m extension cord cross-sectional area
4*6 mm²	4*10 mm²

- Do not use the equipment with damaged high-pressure hoses.
- Position the equipment as close to the water source as possible.
- Discard packaging materials at designated recycling locations in accordance with local laws and regulations.
- Only use accessories and parts authorized by the manufacturer to ensure safe and trouble-free operation.
- Modifying or altering safety valve parameters is strictly prohibited. Explosion hazard.
- Altering the original diameter of the nozzle is strictly prohibited.
- Do not lock the trigger in the operating position.
- Check if the nameplate is securely attached to the equipment. If not, contact your dealer. Never use equipment without a nameplate, as it is unidentifiable and may pose potential hazards. Risk of accident.
- It is strictly prohibited to leave the equipment unattended.
- All power contact points must be waterproof.

- Always unplug the power cord before performing any work on the equipment. Accidental startup hazard.
- Before pulling the trigger, firmly grip the firearm to counteract recoil. Risk of injury.
- According to the DIN 1988, if a check valve is installed in the water supply line, the equipment must be installed on the main drinking water line. Check that with the local water supply company.
- Maintenance and repair of electrical components must be performed by qualified safety personnel. Risk of accident.
- Release residual pressure before disconnecting the hose. Risk of injury.
- Every time before using the equipment, check it for loose and damaged parts. Risk of accident.
- Use only cleaning agents that do not harm the high-pressure hoses or the power cord sheath.

7. Personal Protective Equipment

Operators and other personnel working nearby must wear the following protective equipment and clothing:

- Eye protection—Operators must wear face shields and goggles to protect against water splashes and flying debris. Wear appropriate goggles and a face shield, both of which protect the eyes and face from injury during high-pressure water cleaning.
- Head protection—All personnel in the work area must wear helmets.

  Helmet material must be capable of withstanding mechanical impact without damage. (10G at 8 m/s)
- Hand protection—Operators must wear cut-resistant gloves at all times. The glove lining must be tightly constructed to prevent water penetration.
- Foot protection—Safety shoes must have a steel toe cap with a minimum thickness of 0.5mm to protect the toes. The toe cap must cover at least 30% of the shoe's total length. The shoes must also include a metal metatarsal guard. Safety boots are available in various designs. Purchase them as needed.
- Hearing protection—Operators and others exposed to noise levels exceeding 90 decibels for more than 1 hour must wear appropriate hearing protection. Earplugs or earmuffs are typically essential.
- Body protection—Waterproof clothing can only protect the operator from water splashes and flying debris. It cannot deflect or block direct high-pressure water impact. Therefore, operators must exercise caution to avoid pointing the high-pressure gun at themselves or others nearby during operation.

8. Emergency Medical Treatment

When operating the equipment, in the event of an accident, take the injured person to the hospital immediately.

Characteristics of High-Pressure Water Jet Injuries and Essential Treatment Guidelines:

To minimize the scope of harm and the damage in the event of a water jet injury accident, and to ensure employee safety and health, note the following:

The extent of injury caused by water jet impact varies depending on factors such as the distance and angle between the jet and the human body, the duration of exposure, the nozzle orifice diameter, the pressure level of the jet, and the nature of the surface contaminants being cleaned.

- A water jet perpendicular to the human body at a distance of 200 mm with a pressure exceeding 15 MPa will cause deep-penetrating punctures and ingress of contaminated water into the body.
- A water jet perpendicular to the human body at a distance of 500 mm with a pressure exceeding 15 MPa will cause deep-penetrating punctures and ingress of contaminated water into the body.
- When the water jet strikes the human body at a certain angle, it can create both deep-penetrating puncture wounds and open, eroded wounds.
- When the jet stream "sweeps" or "scrapes" across the human body, it creates an open, eroded wound.
- The risk and severity of injection injuries increase dramatically with higher pressure, larger nozzle size, longer exposure duration, and closer proximity of the jet to the body.
- The full extent of injuries caused by high-pressure water jets, especially internal damage, may not be immediately visible. The depth of penetration is often difficult to determine—superficial wounds may appear minor or even bloodless, yet significant amounts of contaminated water can infiltrate through tiny openings, penetrating skin, muscle, and deeper tissues.
- Since the cleaning water contains bacteria and dirt from the object being cleaned, the jet stream may carry those harmful substances to deeper tissues when penetrating the human body. Therefore, medical personnel must clean wounds thoroughly to avoid infection, and if necessary, incise the wounds.
- When the jet pressure exceeds 100MPa and causes injury to the human body, a thorough examination must be conducted to determine whether deep tissues and bones are affected.
- In the event of a water jet injury, take the injured person immediately to the hospital, and communicate the specific details of the injury to the doctor as thoroughly as possible, especially the penetrating nature of water jet injuries and the risk of contamination or bacterial infection.
- On-site first aid for jet stream injuries: Cover the wound with a clean dressing immediately to stop bleeding and monitor the injured person until medical examination is ready.

9. Unpacking

Unpack the equipment and check the received items against the packing list:

- Main body of the equipment
- Power cord (connected to the equipment)
- High-pressure hose
- Gun handle/Gun lance/Gun tip
- Operation Manual/Spare Parts Manual/Technical documentation
- Final Assembly

10. Optional Accessories

- Rotating nozzle
- High-pressure extension tubes are available in lengths ranging from 10 to 30 meters
- Industrial drain cleaning tools (1 front, 3 rear water jet pipe cleaning nozzles)
- Industrial sandblaster

11. Installation/Configuration

1. Install the high-pressure pump: Place the pump on a well-ventilated, level surface and secure it in place.

2. Attach the gun lance to the gun handle.

3. Attach the gun tip to the gun lance.

4. Connect the high-pressure hose to the gun handle.

5. Connect the high-pressure hose to the outlet end of the high-pressure pump head and tighten the hose connector. Avoid driving over high-pressure hoses.

6. Supply tap water to the equipment and connect the inlet hose to the water inlet.

7. Connect the power cable to a three-phase power supply. Before connecting the power supply, check its voltage and frequency against the nameplate to ensure the power supply is compatible. The equipment must be connected to a properly grounded outlet. An additional safety circuit breaker (30 mA) will increase user safety.

8. Press the power button. After the equipment runs for 2 to 3 seconds automatically, it is ready for use. Firmly grip the gun handle and lance handle with both hands, then pull the trigger to begin operation (before initial use, remove the nozzle and spray water for 2 to 3 minutes to remove any foreign particles from the system).

9. After each high-pressure operation and before turning off the equipment, always pull the gun trigger to release the system pressure.

If it is necessary to remove accessories such as the inlet hose, high-pressure hose, or gun lance, ensure their connectors are thoroughly cleaned and then sealed with clean plastic film or bags to prevent sand, stones, or other foreign objects from entering, which could damage the water pump, gun handle, and nozzle during subsequent use.

Notes:

- Make sure your power supply is compatible with the motor.
- Do not let the equipment run for more than 10 minutes without pulling the gun trigger (the pressure regulating valve is in bypass return water state). High water pump temperature can cause severe damage to the pump packing/seals.
- When using a three-phase motor for the first time, check the phase wire connections to ensure the fan rotates clockwise.
- When you use seawater as the water source for the equipment, wash the equipment after use by flushing fresh water through the pump for at least 10 minutes. (Note: Using non-fresh water as the pump medium will shorten the packing life.)
- To prevent damage to the high-pressure hose and power cord, avoid rubbing them against sharp objects and prevent compression or crushing by ensuring proper placement.
- Ensure the inlet water temperature is below 60°C (140°F).
- Connect the water inlet hose to the inlet port. The inner diameter of the hose must be at least 13 mm (1/2 inch), and the water source pressure must be equal to or greater than the outlet pressure of the equipment. The water inlet depth must not exceed 60, and the inlet pressure must not be less than 10 bar.
- The equipment must only be used with clean water. Using unfiltered water or corrosive chemical solvents may damage the equipment.

12. Operation

1. Turn the faucet on completely.
2. Pull the trigger for a few seconds to release air and relieve pressure in the line.
3. Hold down the trigger and press the switch to start the motor.

Notes:

- Always keep the trigger pressed before starting the motor.
- For models with a pressure switch:

  When the trigger is released, the pressure switch automatically shuts off the motor.

  When the trigger is pulled, the pressure drops and the motor starts automatically, then the pressure is restored. There is a very brief delay in between. Once the trigger is released, do not pull it again within 4/5 seconds. Do not leave the equipment in auto-shutdown mode for more than 10 minutes.

13. Storage

1. Turn off the washer.
2. Turn off the faucet.
3. Release residual pressure until no water flows from the nozzle.
4. Engage the trigger lock.
5. Disconnect the power cord.
6. In winter, let the equipment draw in non-corrosive and low-toxicity anti-freezing fluid before storage. Follow the steps below to prevent freezing and cracking:

   6.1. Remove the inlet hose and high-pressure outlet pipe from the equipment.

   6.2. Run the equipment for 30 seconds to drain the remaining water from the pump and boiler coil (applicable to hot water washers).

   6.3. Use an air pump to remove residual moisture from the high-pressure hose.

   6.4. Fill the equipment with antifreeze.

14. Safety in Maintenance and Repair

14.1. Personal Protection

- Wear safety goggles, safety shoes, and a helmet when working.
- Do not wear loose or torn clothing.
- Remove all jewelry before work.

14.2. Site Environment

- Remove any loose items from the work area, such as trash and oily rags.
- Do not engage in or allow horseplay in the work area.
- Ensure that after completing equipment maintenance, all tools, parts, and rags are removed from any rotating parts.

14.3. Preparation before Maintenance and Repair

- Maintenance personnel must observe all safety labels attached to the equipment.
- Review all safety information related to the work you are about to perform.
- Safety Starts with You. Ensure all equipment is properly shut down.
- Always use tools in good condition.

  Ensure you understand how to use these tools before performing any preventive maintenance work.
- Prepare suitable containers to collect and store fluids and disassembled parts.

- Repeatedly check that all pressure in the system has been released before starting repair work.
- If the equipment must be operated to verify the intended repairs, ensure you know how to stop it.

14.4. Safety Precautions

Stay alert

- To avoid burns, be cautious of hot components that have just been turned off.
- Release all high pressure before removing or disconnecting any lines, parts, or related items. Be alert to potential residual pressure when disconnecting any device from a high-pressure system.
- Do not work on anything supported solely by a jack or crane. Before performing any work, support the equipment with blocks or suitable supports.
- Avoid personal injury. When lifting components weighing more than 25 kg, use a chain hoist or request assistance. Ensure all lifting components (chains, hooks, or hoists) are in good condition and properly rated for the load. Ensure the hook is positioned correctly. Always use a support pole when necessary. The lifting hook should not be loaded from one side only.
- Do not attempt to perform any maintenance or cleaning around the equipment while it is in operation.

14.5. Work Guidelines

- Do not make any unauthorized modifications to the equipment or its components. Use only genuine spare parts manufactured by Fussen.
- All electronic components connected to this system must be maintained in accordance with approved procedures. Never bypass fuses, fuse holders, or circuit breakers with jumper wires.
- Maintain preventive maintenance intervals to ensure proper operation of the equipment.

14.6. High-Pressure Operation

- After the high-pressure water supply is shut off, high-pressure water can remain in the system for an extended period of time. Ensure all water is drained from the equipment before maintaining any part of the system.
- Some accessories, connectors, or spare parts are equipped with weep holes. Do not touch these weep holes with bare hands or attempt to block them to stop water discharge.
- When disassembling any high-pressure connectors, gradually loosen the fastening nut to release any trapped pressure through the weep hole.
- Do not move the guard of the high-pressure hose. If the guard is moved for maintenance, ensure it is reinstalled in its original position before starting the equipment.
- Do not stand on or lean against the high-pressure water hose. You may break the connector, resulting in water leakage.

15. Routine Maintenance

1. Clean the filter of the equipment.
2. Check the sight glass on the water pump.
3. Check for wear at the high-pressure hose connections.
4. Check the high-pressure hose for any damage.
5. Check for any damage to the power cord.

Caution: Before performing any work on the equipment, unplug the power cord from the outlet.

6. Lubrication

Replace the crankcase lubricating oil (oil specification: 15W40) after the first 100 hours of use. Thereafter, replace the lubricating oil every 300 hours of operation.

Steps:

1. Turn off the equipment.
2. Remove the hex nut (drain port) on the crankshaft case to drain the used oil
3. Refill the case with new lubricating oil. Observe through the sight glass until the oil level reaches the red mark on the tank.

Caution: Waste oil can pollute water sources and the environment. Handle it with care.

7. Cleaning

Clean with water and a cloth. Do not use strong chemicals or high-pressure water.

Regularly clean the rear cooling fan cover of the motor to ensure sufficient cooling airflow passes through the motor.

16. Troubleshooting of the High-Pressure Plunger Pump

Symptom	Cause	Remedies
The pump is running but not pumping water.	There is a problem with the water supply.	Turn on the water supply valve; Check if the water supply line is folded; if the water inlet pressure is too low, install a booster pump. Do not allow air to enter the water supply line.
The pump is running, but there is no pressure.	The water supply pipe is clogged.	Inspect the inlet pipeline.
	Air enters the pump.	Check the water supply line or each water inlet connection.
	The one-way valve is malfunctioning or foreign matter has entered.	Clean or replace the one-way valve.
	The pressure regulating valve is malfunctioning or foreign matter has entered.	Clean or replace the pressure regulating valve.
	The nozzle aperture is too large.	Use a nozzle with a suitable aperture.
	Seal assembly failure	Replace the sealing component.
Sudden vibration during pump operation Unstable pressure/pulse No pressure when the pump is running.	There is air in the pump.	Check the water supply line.
	One-way valve failure or ingress of foreign matters	Clean or replace the one-way valve
	O-ring at the lower end of the one-way valve is damaged.	Replace the O-ring.
	The one-way valve is fully damaged.	Replace the one-way valve.
	Seal wear	Replace the seal assembly.
	The inlet pipe is clogged, obstructing water flow.	Check the system for pipe blockages or air leaks. Ensure the inlet pipe diameter of the high-pressure pump is suitable.
	High-pressure pump running idle	Check if the inlet pipeline is blocked and if the diameter is appropriate.
	The accumulator has no pressure.	Repressurize or replace the accumulator.

Low pressure	Seal wear Pressure relief valve spring broken Nozzle worn or damaged Insufficient flow causes the high -pressure pump to idle Insufficient power	Replace the seal assembly. Replace the spring. Replace the nozzle. Check the inlet pipe at the water inlet. Check the power.
Excessive operating noise	Bearing wear Connecting rod wear High-pressure pump idling or air in the pump Misalignment with the power	Replace the bearing and re-lubricate. Replace the connecting rod. Check if the inlet pipeline is clogged and if the size is correct. Check the power connection.
Excessive crankcase temperature	Incorrect lubricant used Incorrect crankshaft oil level Improper bearing fit	Use a suitable lubricant. Adjust the oil level to proper amount. Adjust the bearing clearance.
Water in the crankcase	Seal wear	Replace the seals.

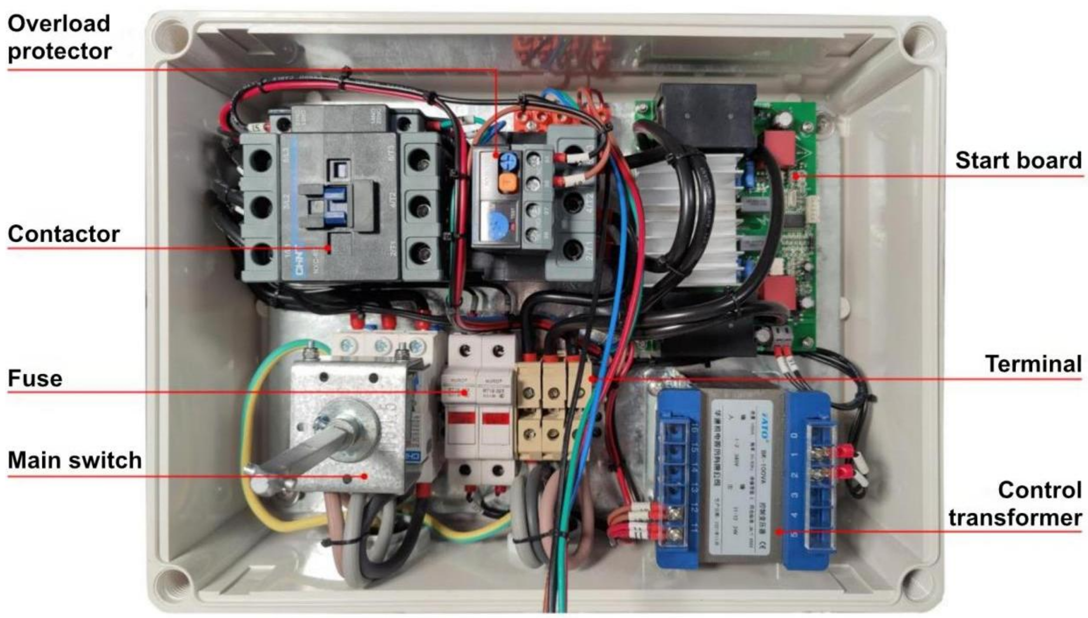
17. Troubleshooting of the Equipment

Symptom	Causes and Remedies
The motor does not start after the start button is pressed.	Check the power supply. Ensure the power cord is properly connected and check it for any damage.
The motor makes noise but won't start.	Voltage too low, phase loss, high-pressure pump blocked. Check the input voltage. Check the voltage junction box. Turn off the equipment and rotate the motor fan blade. If there is any blockage, check the pump.
Pressure drop and unstable equipment performance.	Insufficient water supply. Clean the filter.
When the trigger is released, the system pressure remains high.	The one-way valve in the pressure regulator is stuck or leaking. Clean and inspect the outlet one-way valve of the pressure regulating valve for any sticking or leaks. Replace the O-ring.
When the trigger is operated, the system pressure does not reach the proper working pressure.	Leaking safety valve, high-pressure hose joint, or gun handle; air in the pump; damaged high-pressure nozzle, incorrect setting or wear of pressure relief valve; worn piston gasket. Turn off the equipment. Inspect for leaks, replace the nozzle, reset the pressure or perform repairs, and replace the piston gasket.
When the equipment starts and the pressure reaches approximately two-thirds of the total pressure, the high-pressure water hose vibrates violently.	There is debris in the inlet/outlet valve of the high-pressure pump. Turn off the equipment, remove bolts and valves, clean the debris, and check if the valve movement forms a tight seal with the valve seat.
There is excessive leakage from the water pump.	Gasket damaged Replace the gasket. (Be sure to replace the O-ring)
Excessive noise	There is air in the pump, and one or more valve springs are damaged or have weak elasticity. There is debris in the valve. Damaged crankcase or motor bearing. Check the inlet water pressure, replace the spring, clean the valve, and replace the bearing.
Water in the lubricating oil	The oil seal is damaged. Excessive air humidity (causing condensation inside the crankcase), oil seal is damaged. Inspect or replace the oil seal. Replace the crankcase oil regularly. Replace the seal.

Note:
In the event of a malfunction or other damage not mentioned in the user manual, we recommend contacting your supplier for further repairs or part replacement.

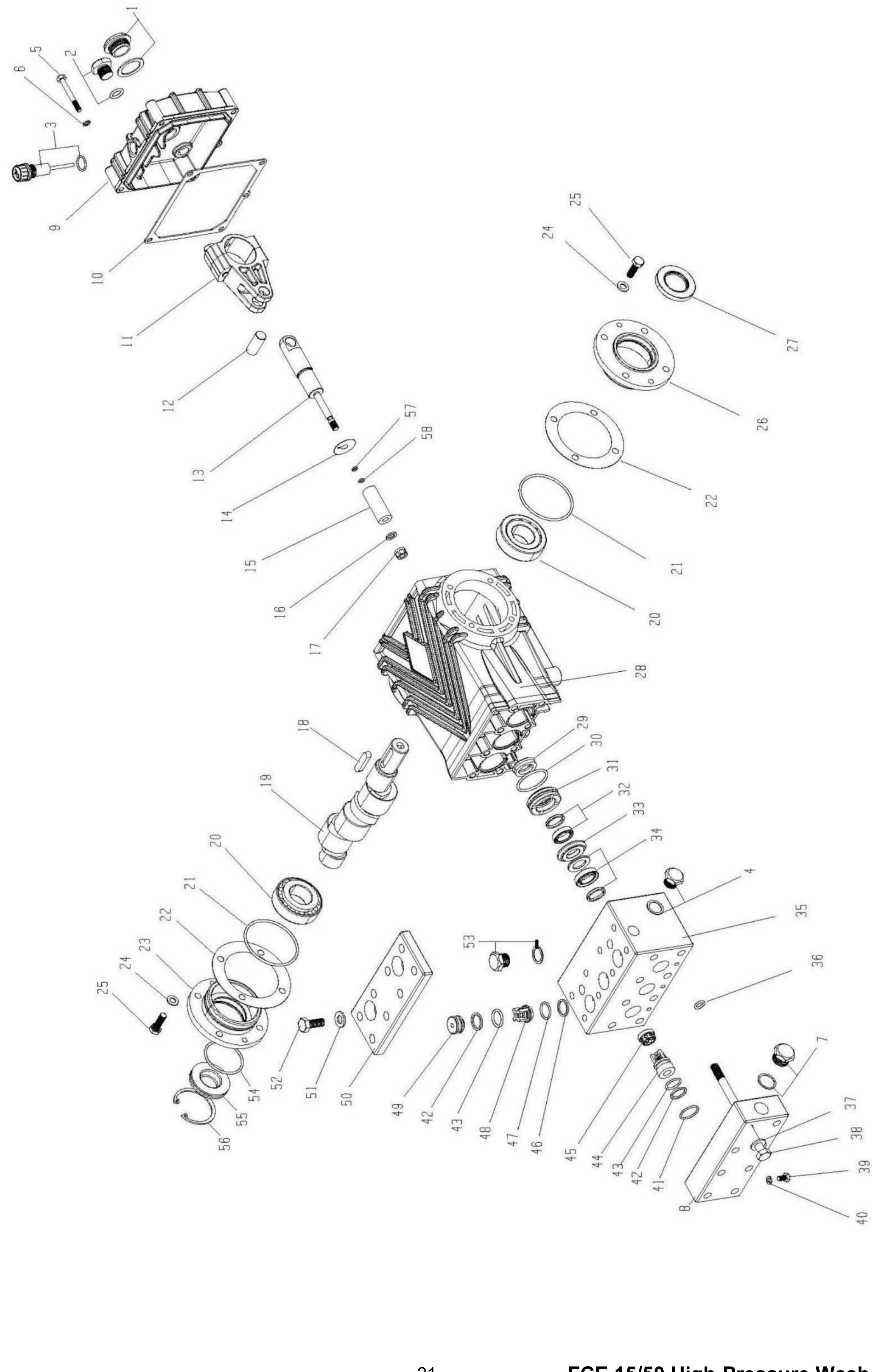
18. Appendix

18.1. List of Components



No.	Material No.	Name	Quantity	Note
1	3.998-005	Control cabinet	1	
2	6.150-001	Low-speed motor	1	
3	3.997-002	Rotomolded water tank	1	Includes float switch
4	6.610-001.2	Safety valve	1	
5	6.600-005.1	Pressure regulating valve	1	
6	7.640-006	Filter	1	
7	7.300-054	Elbow	1	
8	6.660-012	Pressure gauge	1	
9	7.640-009.1	Secondary filter	1	
10	7.300-054	Elbow	1	
11	7.602-002	Water supply pipe	1	
12	7.300-027	Union	1	
13	3.997-003	Tire	4	
14	5.200-002	Thermal overload protector	1	
15	5.200-003	Contactor	1	
16	5.220-002	Soft start circuit board	1	
17	5.210-001	Transformer	1	
18	6.640-013	Gun handle	1	
19	3.100-001.2	Gun lance	1	1.2 m
20	7.300-032	Connector	1	
21	7.210-042	Fan nozzle	1	
22	7.210-037	Straight jet nozzle	1	
23	7.600-023	High-pressure hose	1	Available in 10, 20, or 30 meters, with extendable docking options.

Exploded view of plunger pump



Bill of Materials of plunger pump

No.	Name	Specifications	Code	Qty.	No.	Name	Specifications	Code	Qty.
1	Hexagon oil sight glass with flat washer	G3/4		1	30	O-Ring of bronze guide bushing	NBR70 36.17*2.62	7.906-005.5	3
2	End cap	G3/8+O-ring		1	31	Bronze guide bushing	Φ43*14	7.906-040.1	3
3	Oil dipstick	G3/8+O-ring	7.906-034	1	32	Secondary water seal	15*22*4.5	7.906-007.4	3
4	End cap	G3/8 + Stainless steel bonded seal	7.400-011	1	33	Brass pressure ring	Φ39*6	7.906-042.1	3
5	Hex head screw	M6*50	7.301-008	5	34	Main water seal	15*24*9.3	7.906-007.11	3
6	Double-sided serrated washer	M6	7.906-146	5	35	Pump head	166*89*86	7.906-031	1
7	End cap	G1/2 + Stainless steel bonded seal	7.400-001.2	1	36	Cooling water O-ring	9.9*2.62	7.906-005.1	3
8	Low-pressure cover	166*68*34	7.906-031	1	37	Double-sided serrated washer	M10	7.906-145	8
9	Back cover	148*124*42	7.906-011	1	38	M10 hex head screw	M10*140	7.301-010	8
10	Rubber gasket	Large Gasket 147*123	7.906-010	1	39	M6 hex head screw	M6*10	7.301-011	3
11	Connecting rod	FSC-65	7.906-001.4	3	40	Bonded seal	M6	7.906-046	3
12	Piston pin	14*32	7.906-033	3	41	Low-pressure cover O-ring	NBR70 20.3*2.62	7.906-005.2	3
13	Plunger rod	FS C16/50 22*145.5	7.906-002.1	3	42	Retaining ring	18*22*1.55	7.906-008.1	6
14	Air cap gasket	Φ28*Φ8.5*0.5	7.906-132	3	43	O-ring	NBR70 17.12*2.62	7.906-005.4	6
15	Ceramic tube	Φ 15*Φ8*46 C1650	7.906-003.4	3	44	Low-pressure valve	Φ22*30	7.906-013	3
16	Copper gasket	13*18*1.5	7.906-012.1	3	45	One-way valve copper support	Φ21.5*11.5	7.906-059	3
17	Ceramic tube lock nut	M8 (produced with 13#)	7.906-002.1	3	46	High-pressure valve retaining ring	18*22*1.05	7.906-008	3
18	Flat key	8*7*35	7.906-035.1	1	47	High-pressure valve O-ring	NBR70 17.12*2.62	7.906-005.4	3
19	Crankshaft	FSC 22L solid	7.906-025.8	1	48	High-pressure valve	Φ21.6*19.5	7.906-013.1	3
20	Tapered roller bearing	32206J	7.906-014	2	49	Copper seat for high-pressure valve	Φ22*13.5	7.906-072	3
21	End cap O-ring	NBR70 67.95*2.62	7.906-005.3	2	50	High-pressure valve cover	166*70*15	7.906-031	1
22	Adjusting shim	0.1/0.2/0.5mm	7.906-010.4.5.6	1 each	51	Double-sided serrated washer	M10	7.906-145	8
23	Closed bearing cover	Φ 106*35.8	7.906-011	1	52	304 hex head screw	M10*30	7.301-015	8
24	Double-sided serrated washer	M8	7.906-022	8	53	End cap	G1/4 + bonded seal	7.400-001	2
25	Hex head screw	M8*25	7.301-009	8	54	Oil sight glass sealing ring	44.12*2.62	7.906-049.10	1
26	Open bearing cover	Φ 106*30.8	7.906-011	1	55	Oil sight glass	52*14	7.906-020.2	1
27	Oil seal	TC30*52*7 30*52*7	7.906-006	1	56	Oil sight glass retaining ring	Diameter 50 (for holes)	7.906-043.2	1
28	Crankcase	208*174	7.906-011	1	57	45-degree open washer	5*7.7*1.3	7.906-008.4	3
29	Housing oil seal	SC20*35*7 20*32*7		3	58	Plunger rod O-ring	4.47*1.78	7.906-019	3

18.2. High-Pressure Washer Circuit Diagram